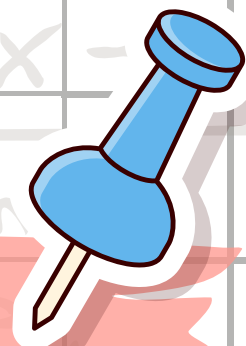


Fraction Funland



$$\frac{1}{4}$$



+

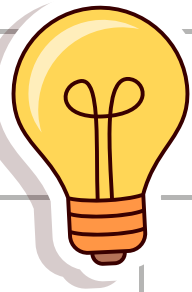


MATHEMATICS 4

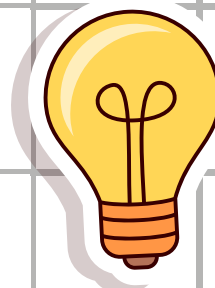
$$\frac{2}{3}$$

3
1
7

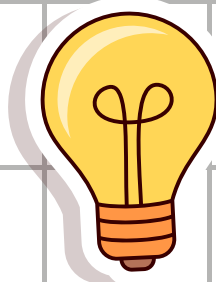
Learning Objectives



Understanding
fractions and their
form



Comparing,
ordering and
equivalency



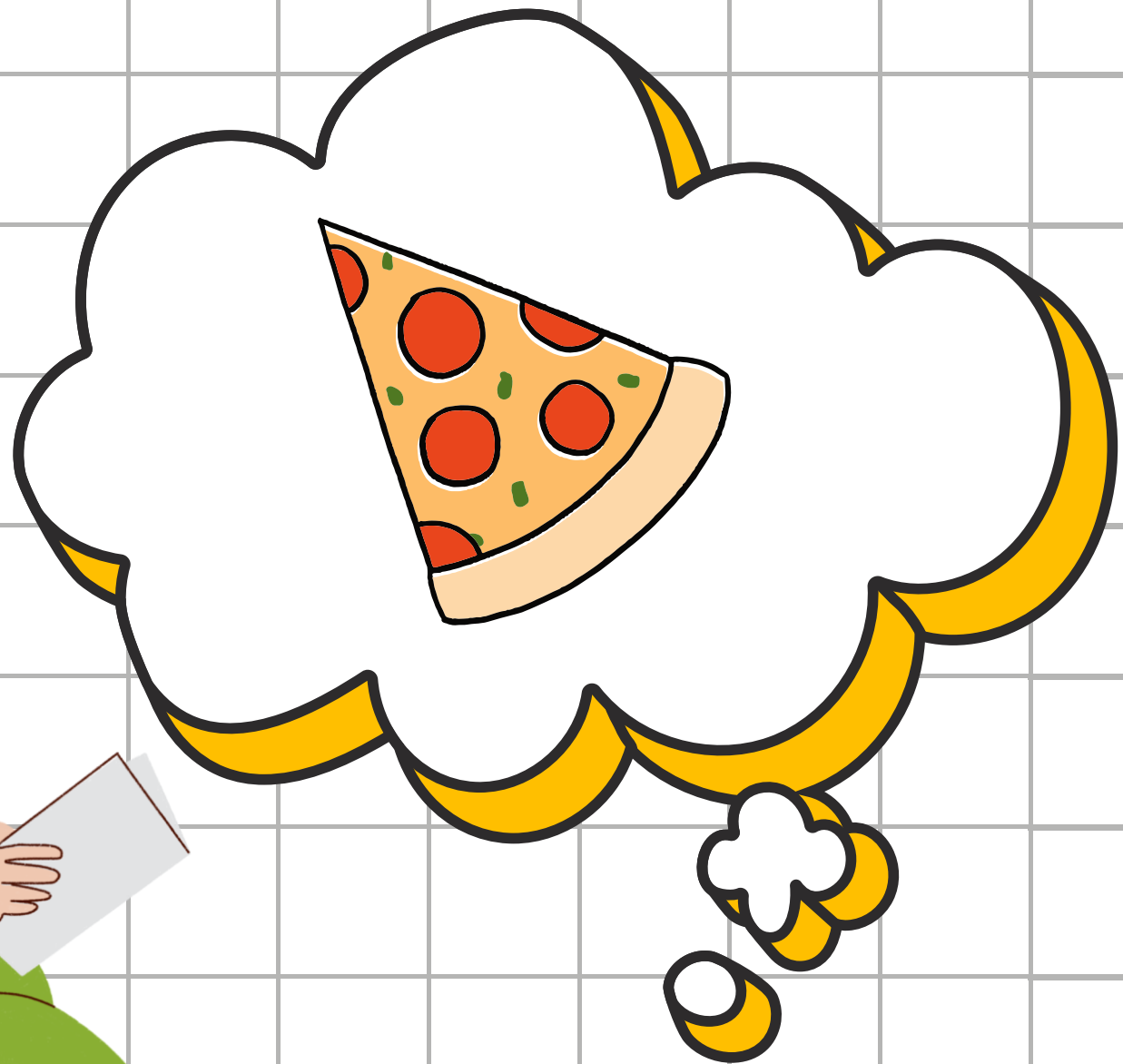
Oprations with
fractions

1
2



Fraction

It represents part of a whole.



numerator

1

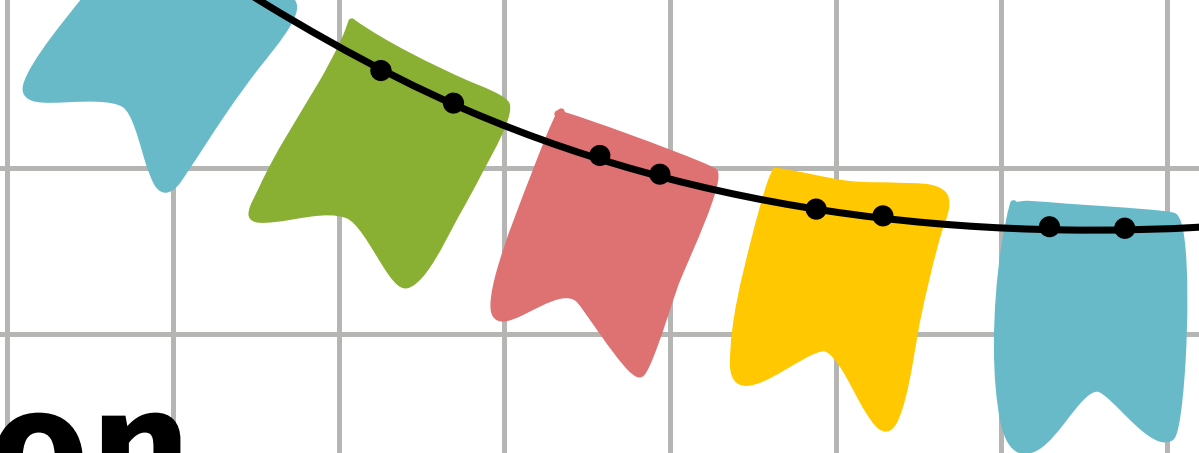


3

denominator



Let's Recall



Types of Fraction

Proper Fractions

The numerator is smaller than the denominator

numerator $\frac{1}{4}$ denominator

Unlike Fractions

The numerator is equal or greater than the denominator

numerator $\frac{8}{3}$ denominator

Mixed Number

a combination of a whole number and a proper fraction

whole number $2\frac{1}{3}$ proper fraction



Let's Recall

Like fractions

fractions that have the same denominators

$$\frac{2}{8} \text{ and } \frac{5}{8}$$

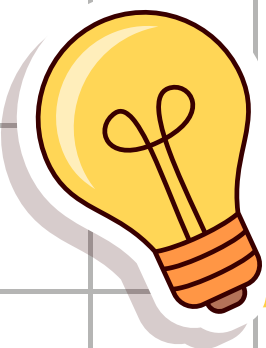
↑ ↑
same denominators

Unlike fractions

fractions that have different denominators

$$\frac{1}{4} \text{ and } \frac{1}{5}$$

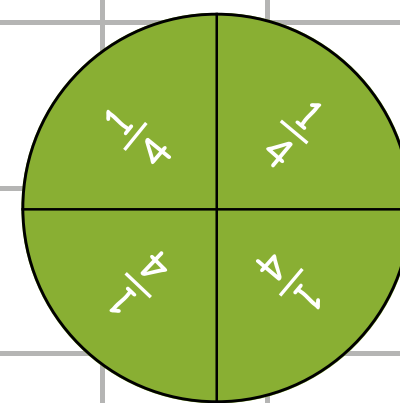
↑ ↑
different denominators



Adding Similar Fractions

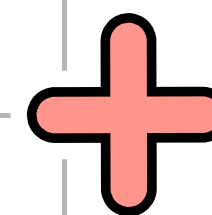
Add the numerators and copy the common denominator.
Lowest term if possible.

$$\frac{4}{10} + \frac{3}{10} = \frac{7}{10}$$



$$\frac{8}{15} + \frac{2}{15} = \frac{10}{15} \div \frac{5}{5} = \frac{2}{3}$$

$$\frac{3}{4}$$





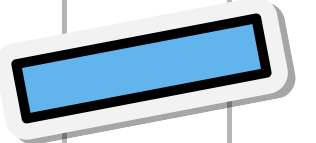
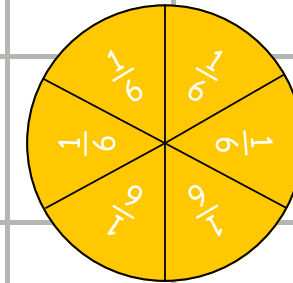
Subtracting Similar Fractions



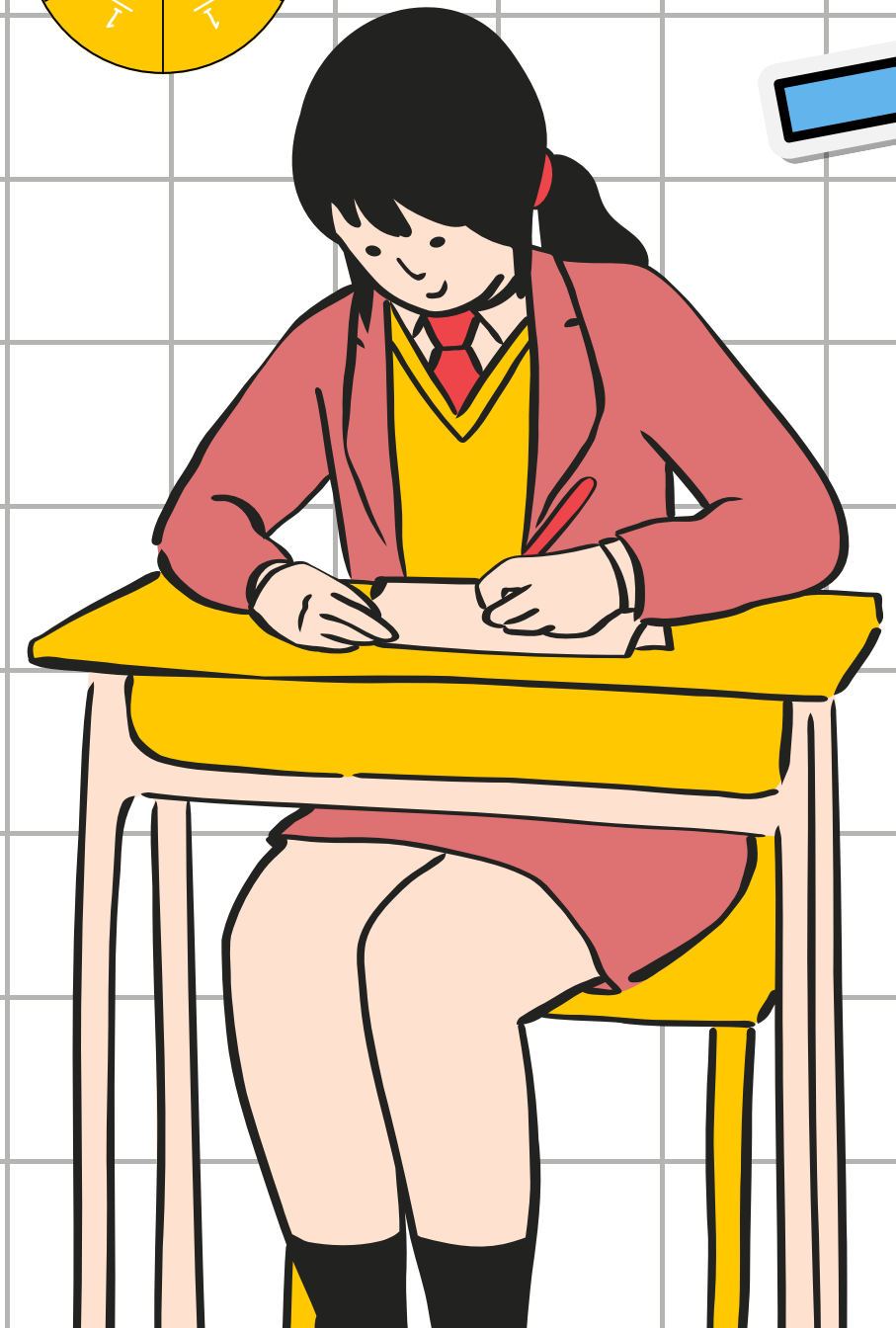
Subtract the numerators and copy the common denominator. Lowest term if possible.

$$\frac{8}{15} - \frac{4}{15} = \frac{4}{15}$$

$$\frac{2}{3}$$



$$\frac{16}{24} - \frac{12}{24} = \frac{4}{24} \div \frac{4}{4} = \frac{1}{6}$$



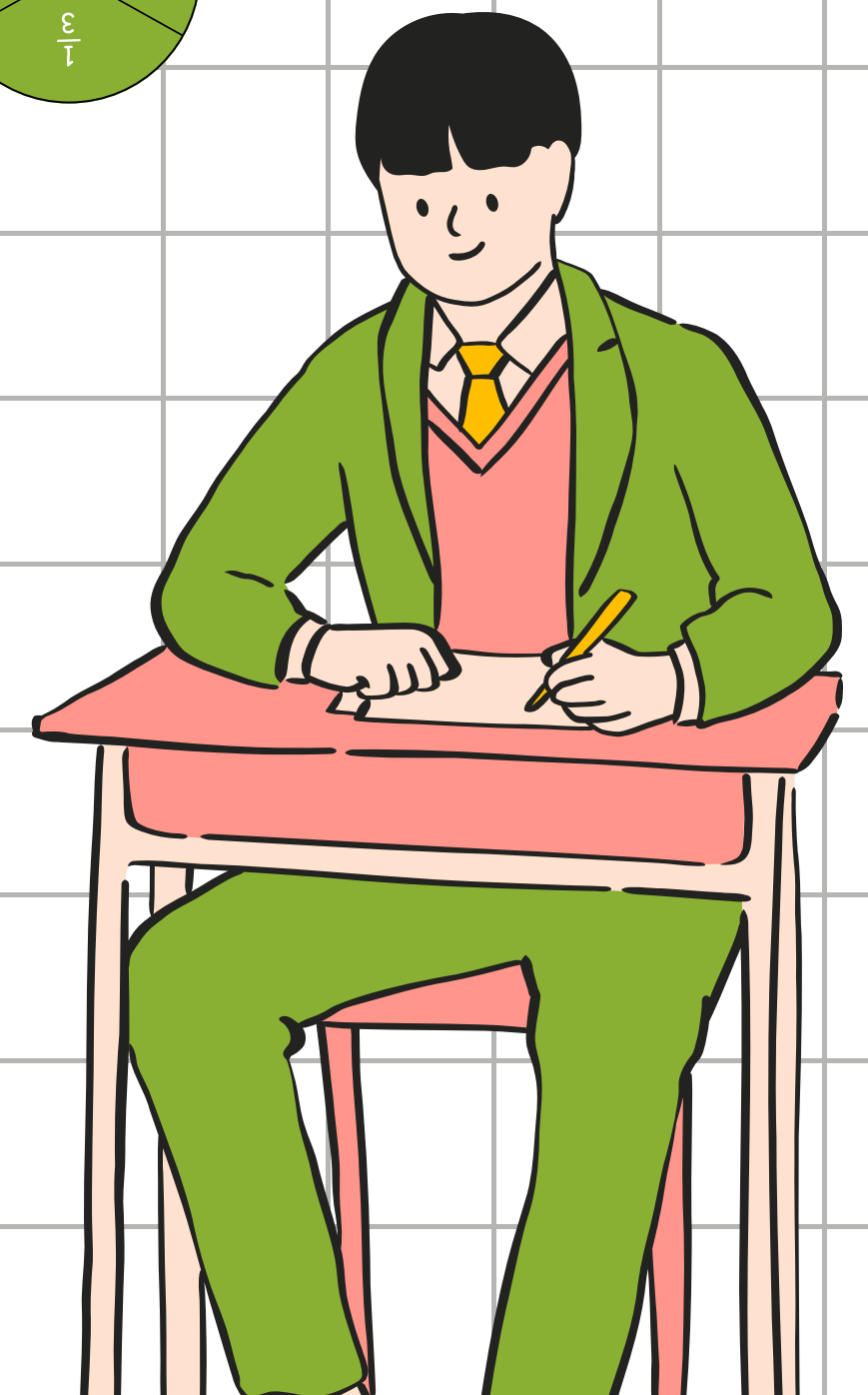
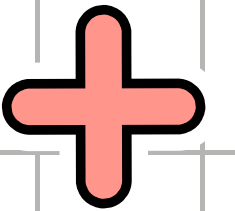
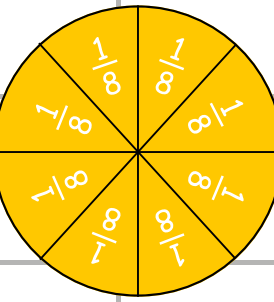
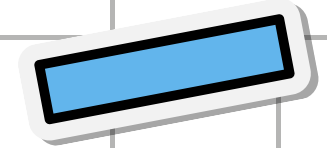
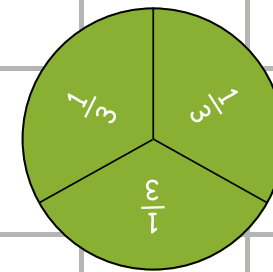


Adding and Subtracting Mixed Numbers

Add/ Subtract the similar fractions then the whole numbers. Lowest term if possible.

$$6\frac{2}{7} + 3\frac{1}{7} = 9\frac{3}{7}$$

$$12\frac{7}{10} - 5\frac{3}{10} = 7\frac{4}{10} \div \frac{2}{2} = 7\frac{2}{5}$$



Activity



$$1.) \frac{5}{8} + \frac{2}{8} =$$

$$2.) \frac{7}{18} - \frac{3}{18} =$$

$$3.) 9\frac{4}{15} - 2\frac{2}{15} =$$

$$4.) 4\frac{3}{12} + 2\frac{1}{12} =$$

$$5.) 10\frac{4}{8} + \frac{2}{8} =$$

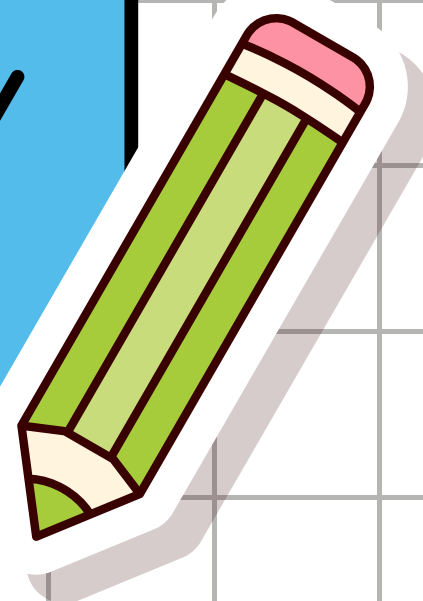
$$6.) \frac{4}{21} + \frac{3}{21} =$$

$$7.) \frac{12}{17} - \frac{9}{17} =$$

$$8.) 8\frac{7}{14} - 5\frac{5}{14} =$$

$$9.) 6\frac{6}{24} + 1\frac{2}{24} =$$

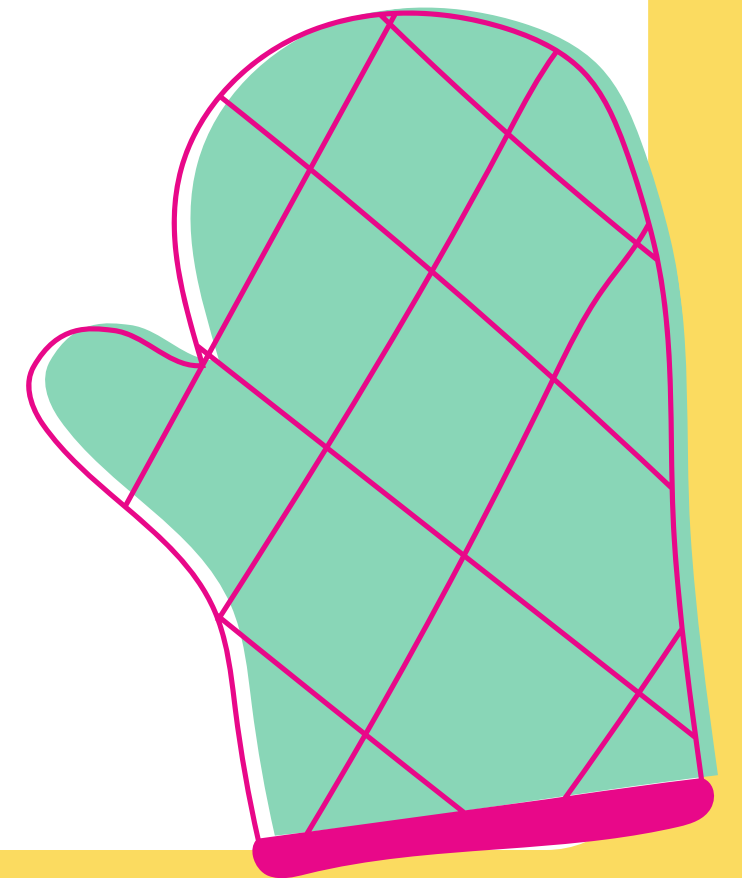
$$10.) 13\frac{8}{25} + \frac{3}{25} =$$



UNLIKE DENOMINATORS

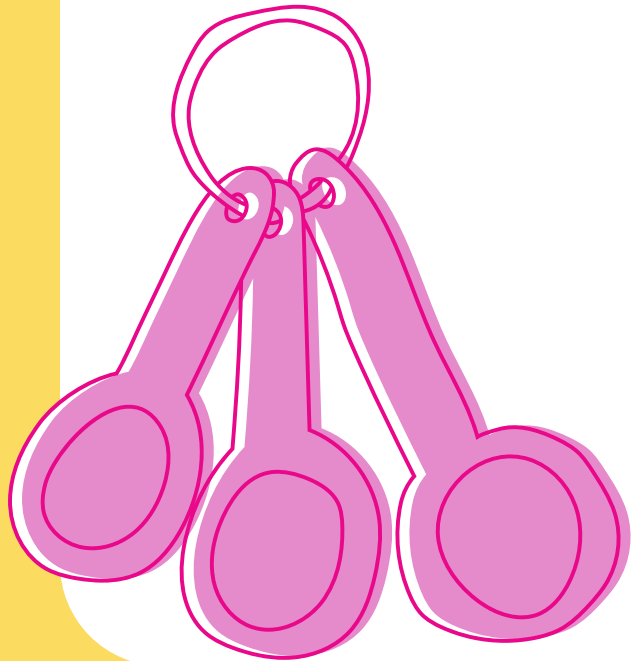
- Unlike denominators are fractions with different denominators.

- Example: $\frac{1}{4}$ and $\frac{2}{3}$



ADDING FRACTIONS WITH UNLIKE DENOMINATORS

- Find a common denominator, also called a least common multiple.
- The least common denominator for these two fractions is 12.
- Rename the fractions with the common denominators.

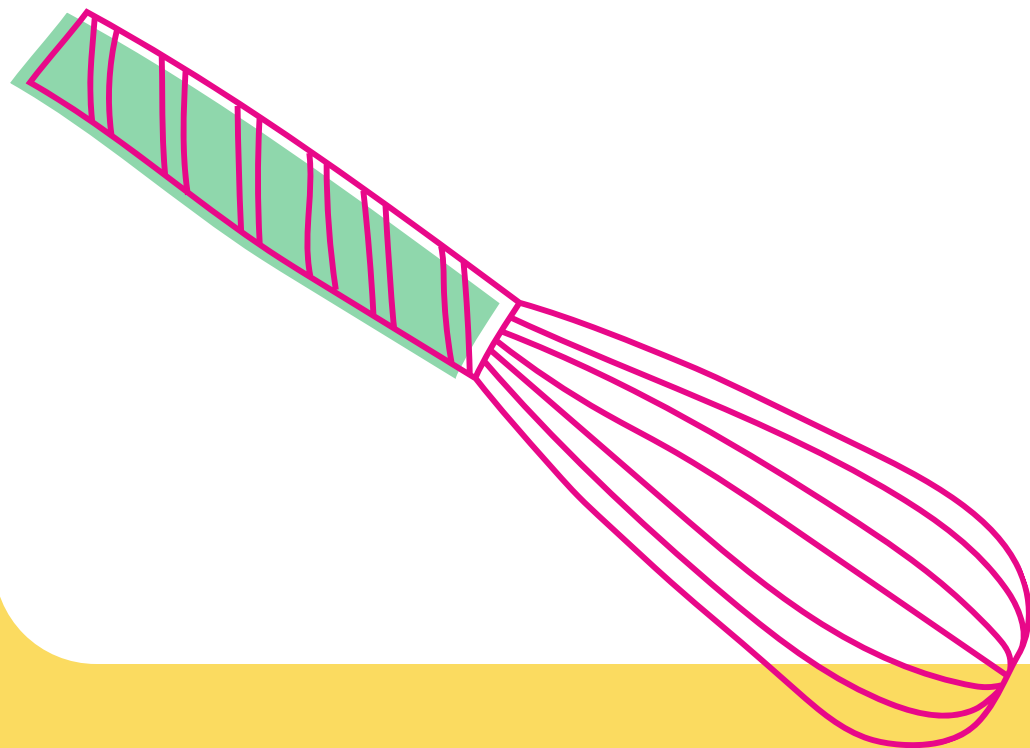


$$\frac{1}{4} = \frac{3}{12}$$

$$\frac{2}{3} = \frac{8}{12}$$

ADDING FRACTIONS WITH UNLIKE DENOMINATORS

- Add the numerators.
- Keep the denominator the same.
- Simplify if needed.



$$\frac{3}{12} + \frac{8}{12} = \frac{11}{12}$$

FIND THE COMMON DENOMINATOR

Name the least common multiple for the fractions below.

$$\frac{1}{2}$$

$$\frac{2}{5}$$

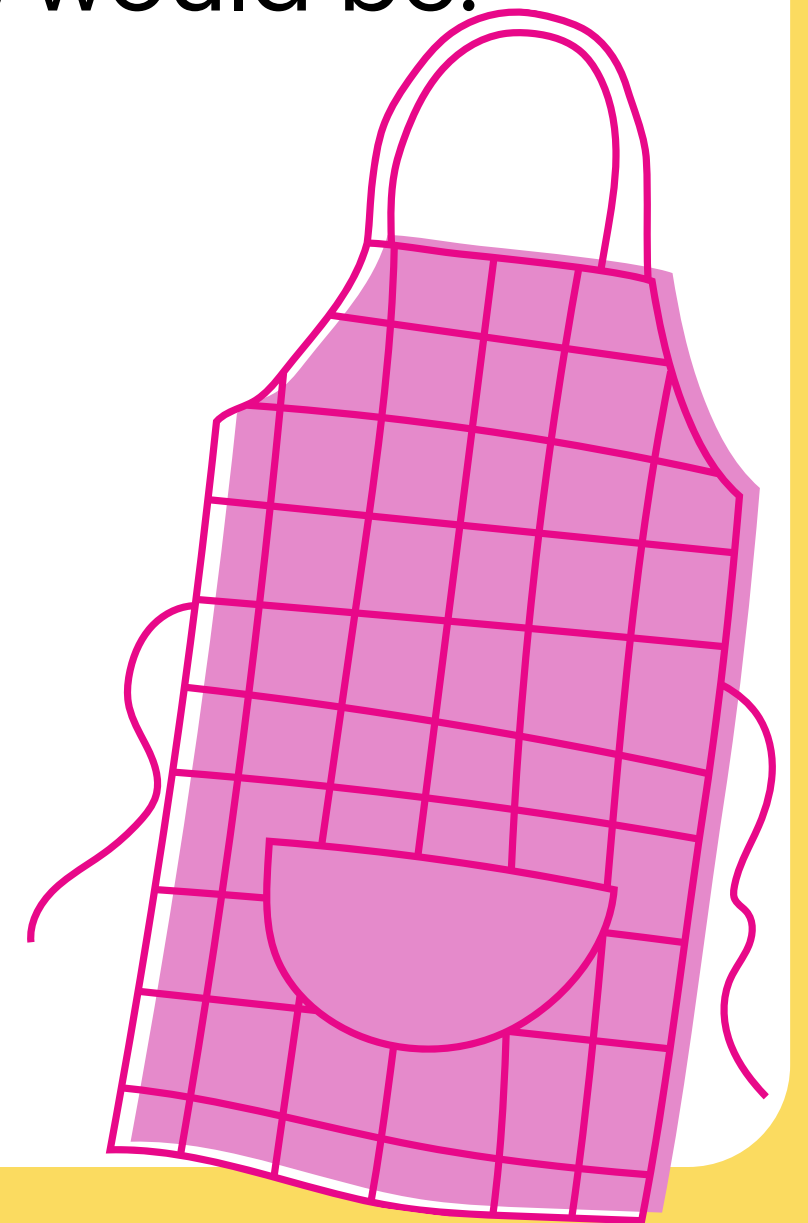


BAKED TO PERFECTION!

Their common denominator is 10. So the fractions would be:

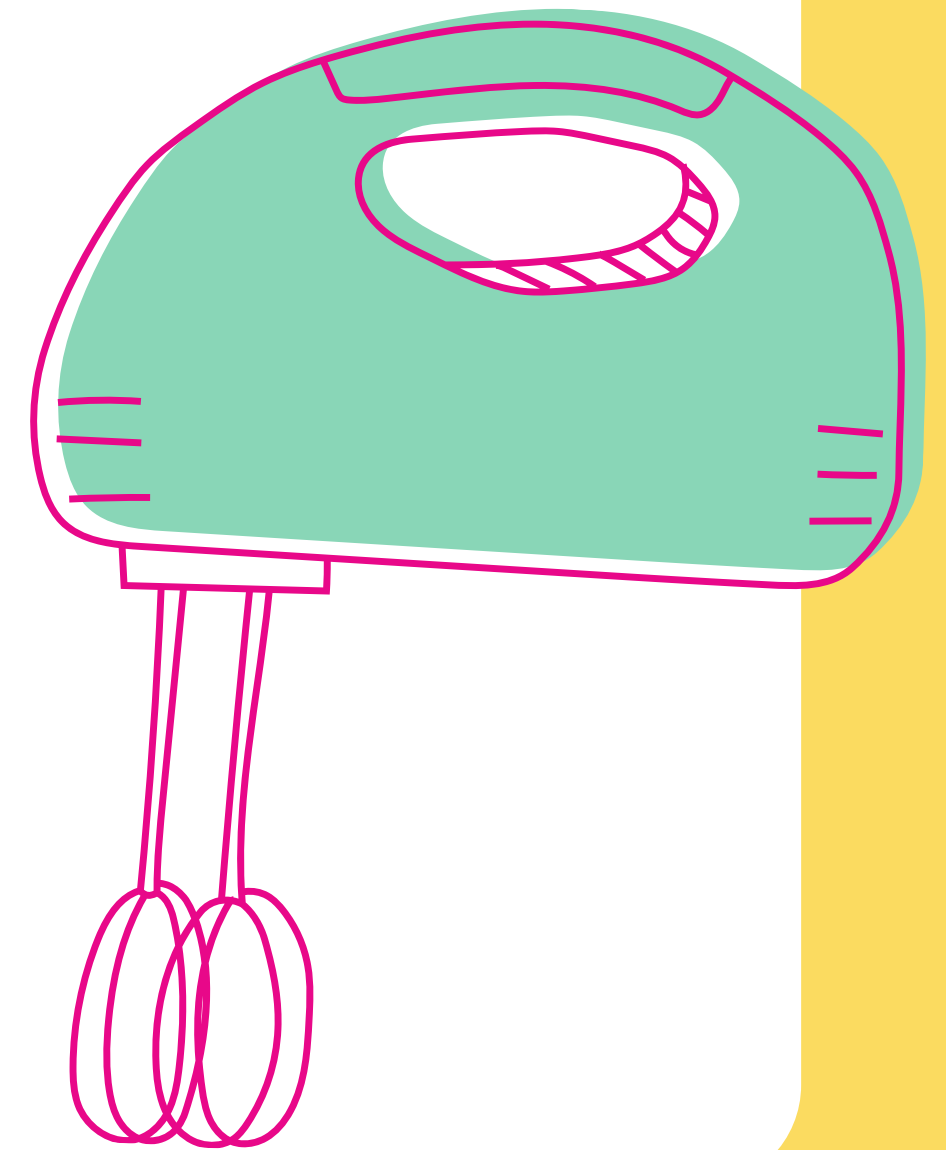
$$\frac{5}{10} \quad \frac{4}{10}$$

Now, add the fractions together!



YOUR RECIPE IS
COMPLETE!

$$\frac{5}{10} + \frac{4}{10} = \frac{9}{10}$$



NOW IT'S YOUR TURN TO WHIP UP A SOLUTION!

Sophia is baking a batch of cupcakes. Her recipe calls for:



$\frac{1}{2}$ cup of sugar and $\frac{2}{3}$ cups of brown sugar

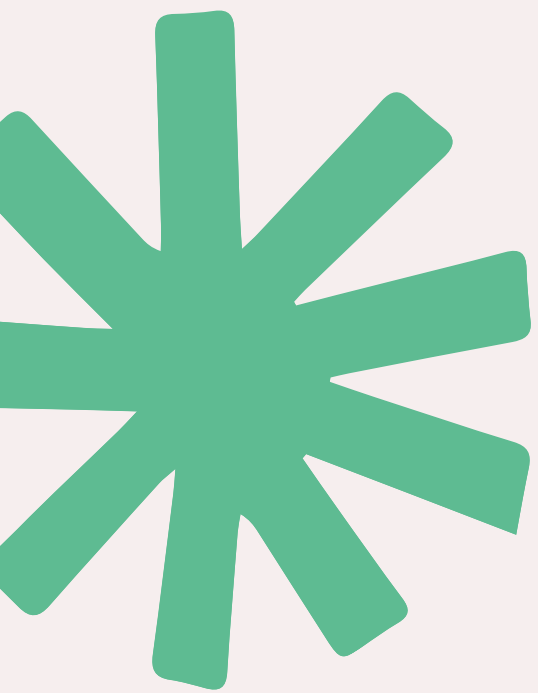
How much sugar will Sophia use in total?

FRESH OUT OF THE OVEN!

Did you get it right?



$$\frac{7}{6} = 1\frac{1}{6}$$



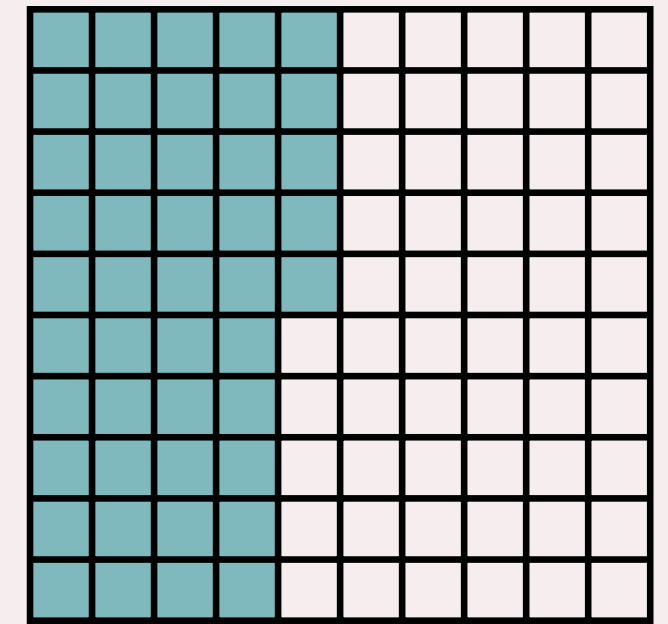
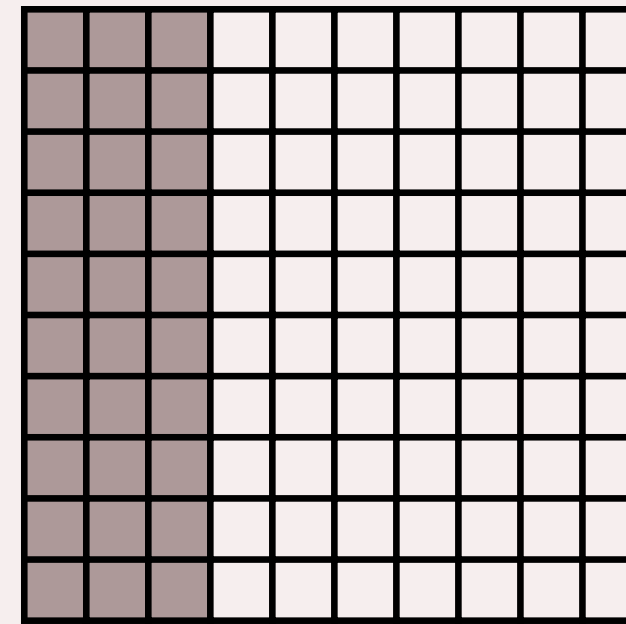
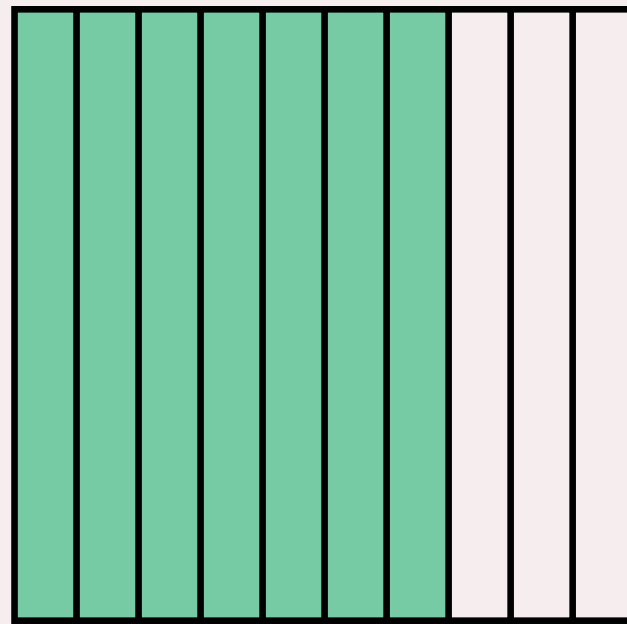
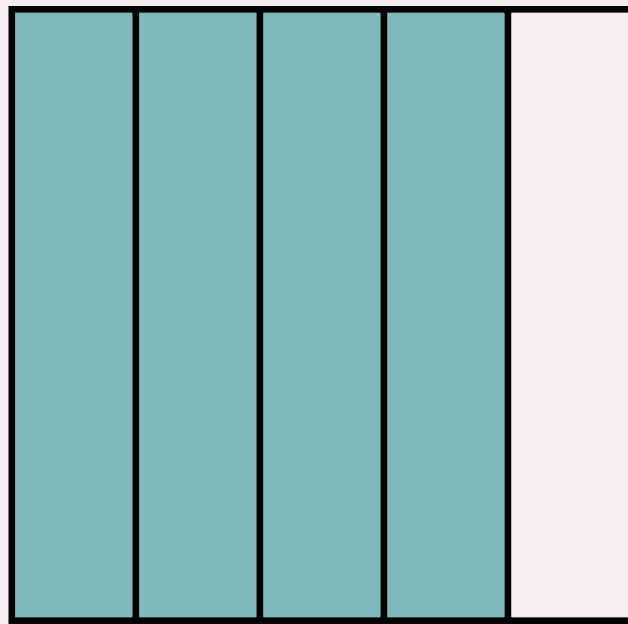
Converting between

Fractions, Decimals and Percentages



Fractions

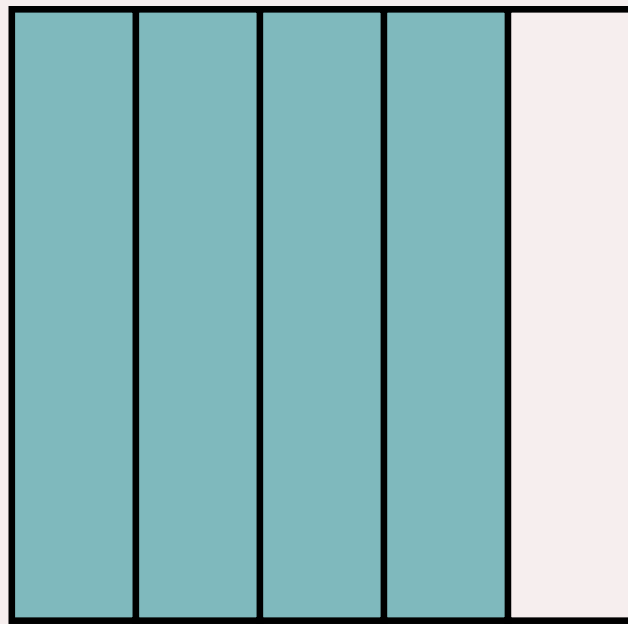
A fraction represents a part of a whole.
Write down the fractions represented by the diagrams:



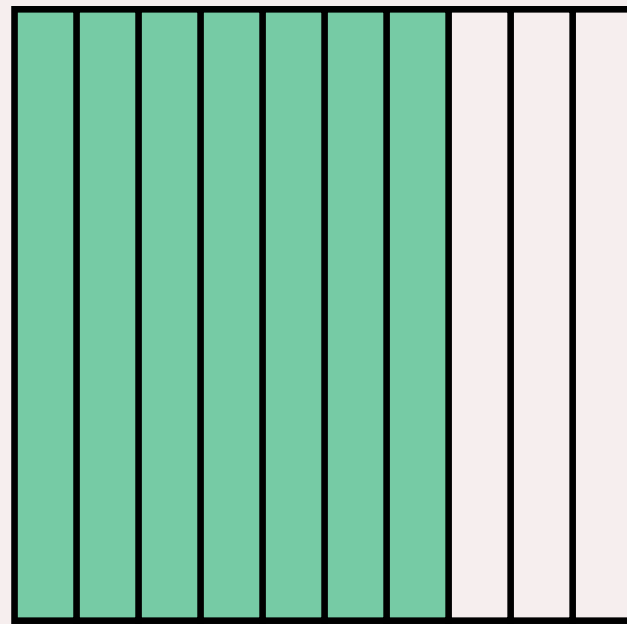
Fractions

Answer Key

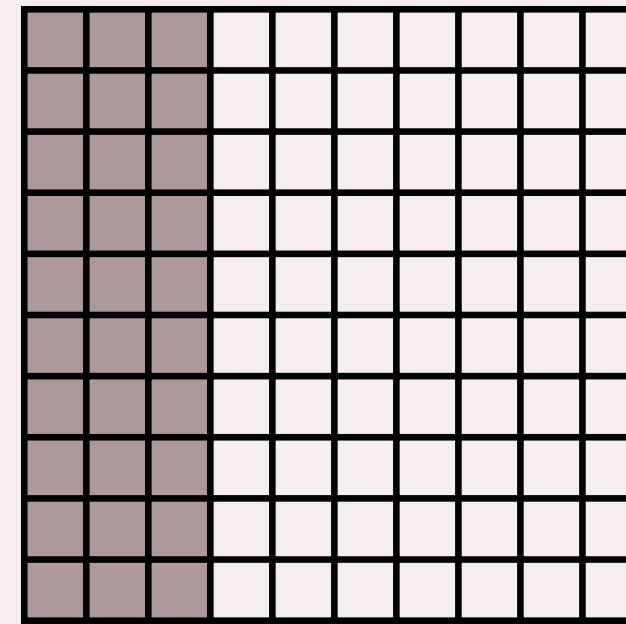
A fraction represents a part of a whole.
Write down the fractions represented by the diagrams:



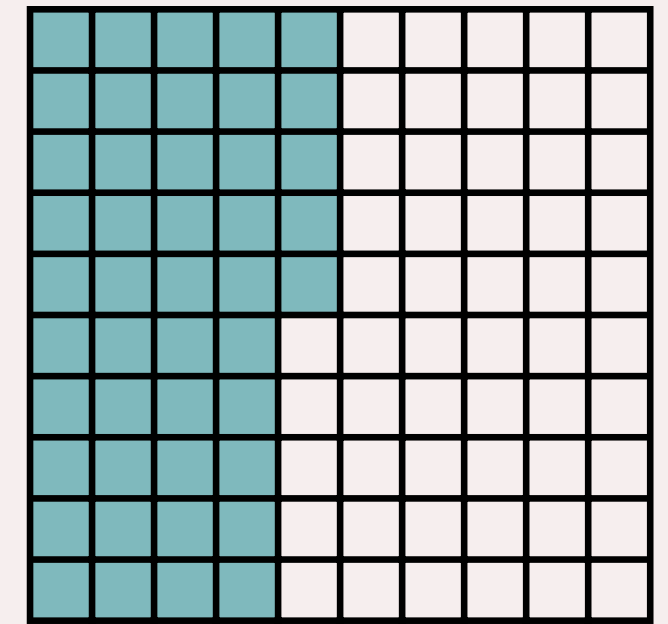
$$\frac{4}{5}$$



$$\frac{7}{10}$$



$$\frac{3}{10}$$

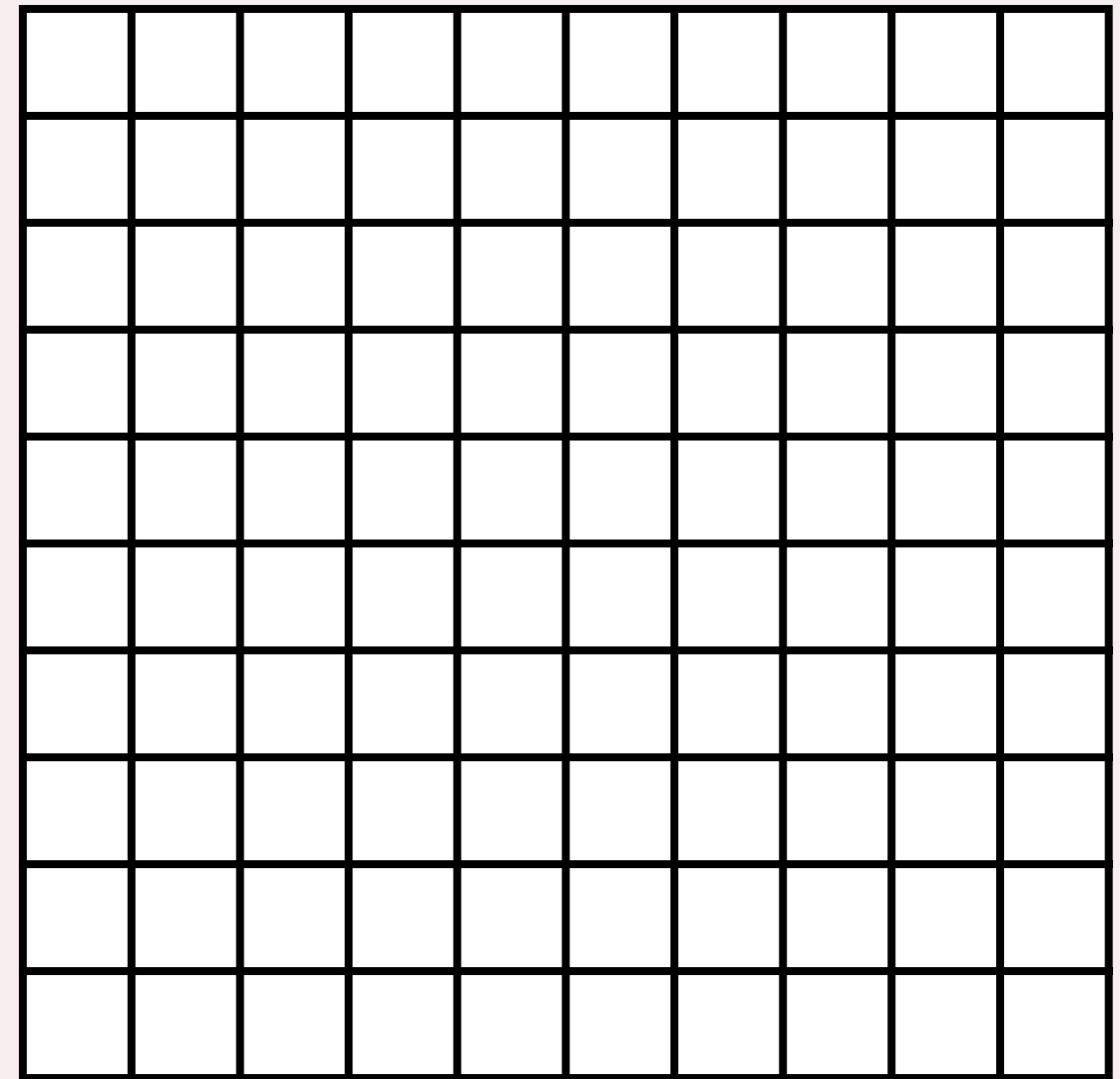
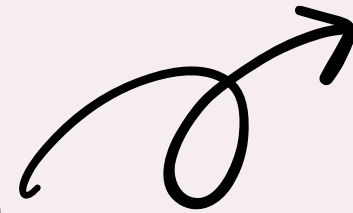


$$\frac{45}{100}$$

Percentages

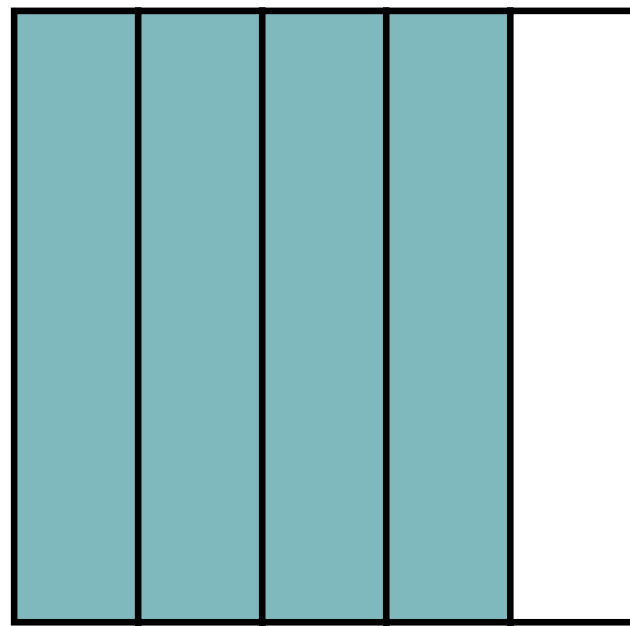
Percentages are a way of expressing a part of a whole as a fraction of 100. They are useful for comparing quantities

Out of 100



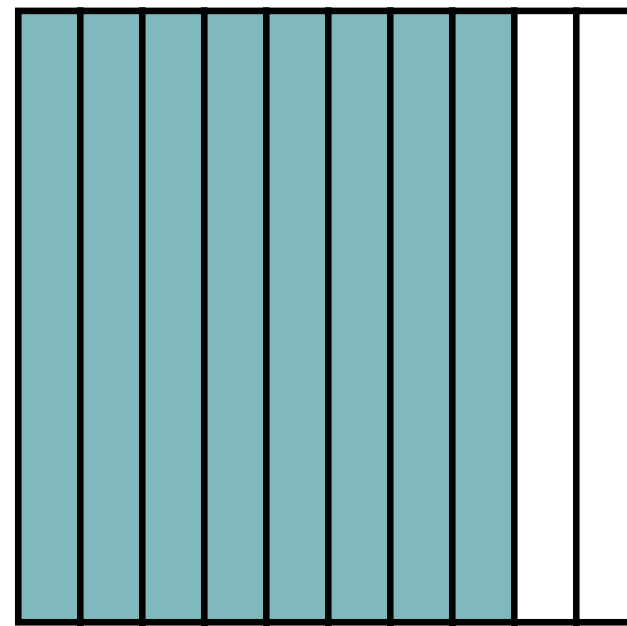
Fractions to Percentages

In order to convert a fraction into a percentage, you need to find the equivalent where the denominator is 100.



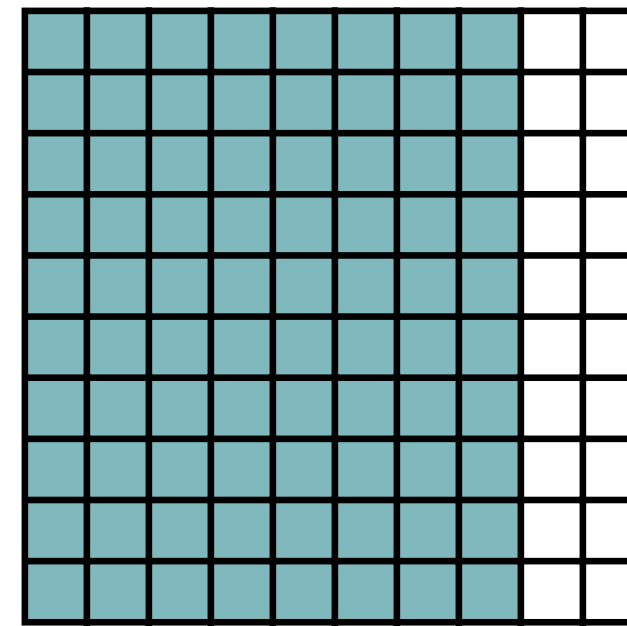
$$\frac{4}{5}$$

=



$$\frac{8}{10}$$

=



$$\frac{80}{100}$$

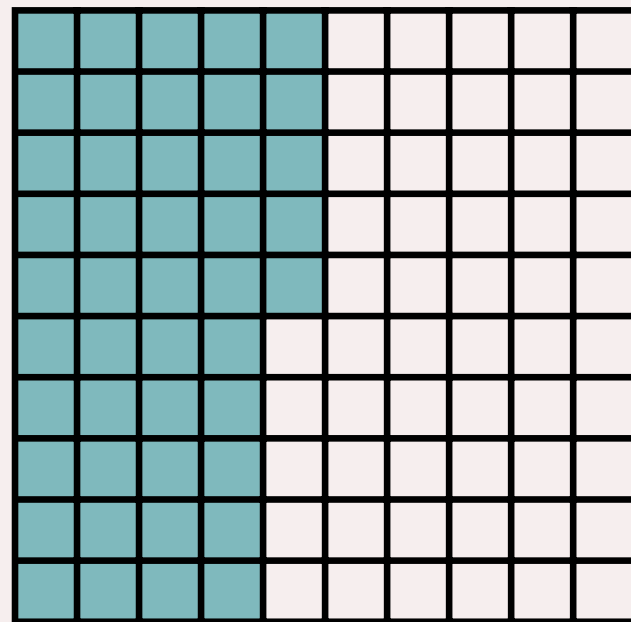
→

80%

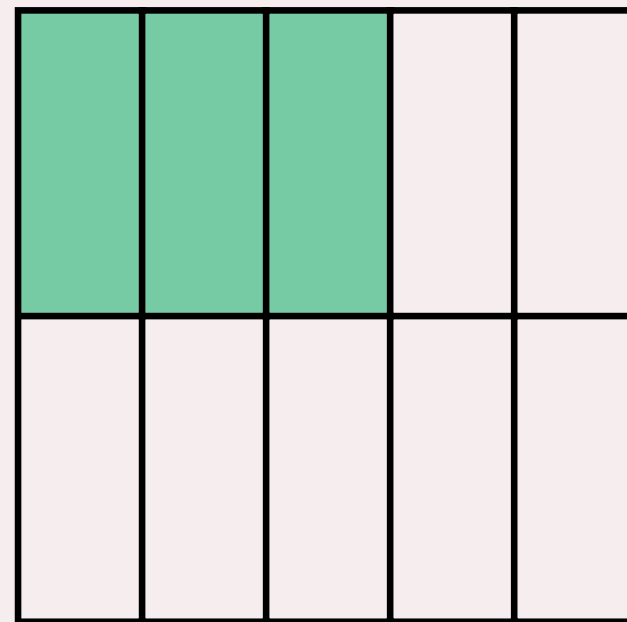
The numerator is the percentage.

Fractions to Percentages

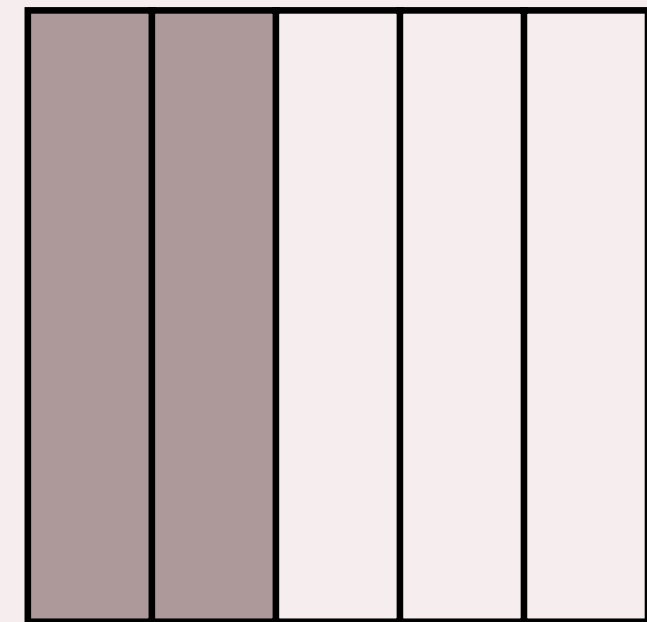
Convert the following fractions into percentages:



$$\frac{45}{100} =$$



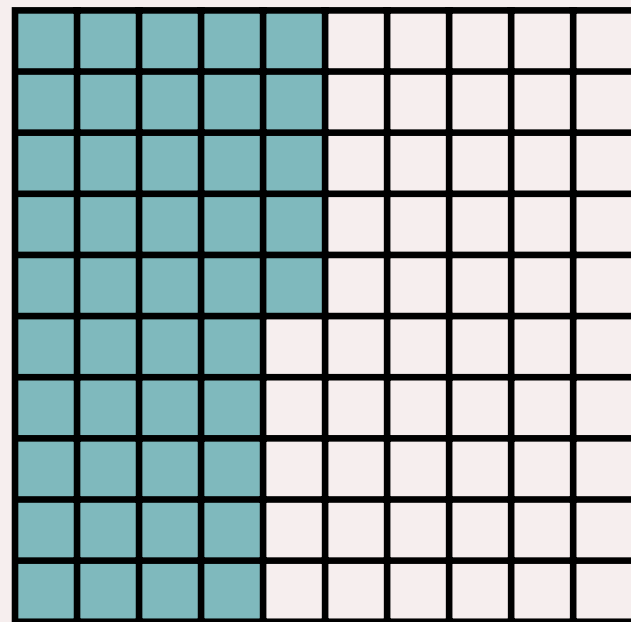
$$\frac{3}{10} =$$



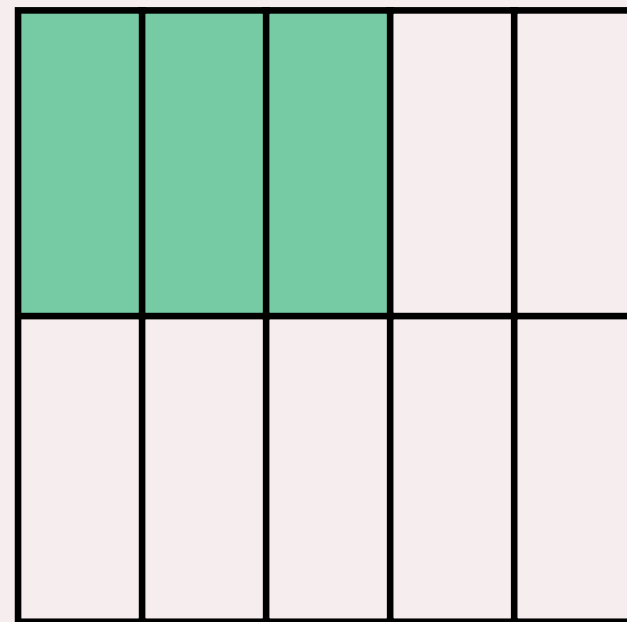
$$\frac{2}{5} =$$

Fractions to Percentages

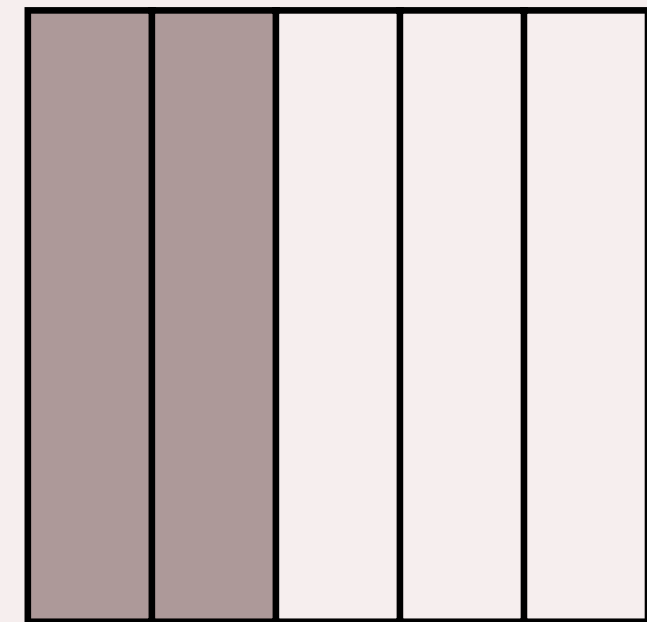
Answer Key: Convert the following fractions into percentages:



$$\frac{45}{100} = 45\%$$



$$\frac{3}{10} = \frac{30}{100} = 30\%$$



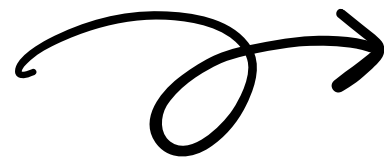
$$\frac{2}{5} = \frac{40}{100} = 40\%$$

Percentages to Decimals

In order to convert percentages into decimals, divide by 100

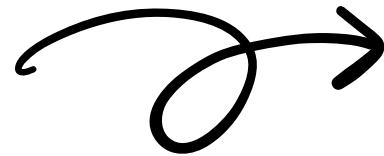
Divide by 100

88%



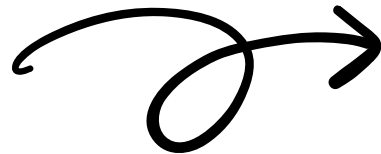
0.88

70%



0.7

7%



0.07

Fractions to Decimals

To convert fractions into decimals,
divide the numerator by the denominator

$$\frac{3}{8} = 3 \div 8 = 0.375$$

$$\begin{array}{r} 0.375 \\ 8 \overline{) 3.000} \end{array}$$

Divide the numerator by the denominator



Let's Learn

MULTIPLYING & DIVIDING FRACTIONS



Let's start with
MULTIPLYING!

$$\frac{2}{3} \times \frac{3}{4} = \frac{2 \times 3}{3 \times 4} = \frac{6}{12} = \frac{1}{2}$$

The diagram illustrates the simplification of the fraction $\frac{6}{12}$ to $\frac{1}{2}$. A curved arrow labeled $\div 6$ points from the numerator 6 to the numerator 1. Another curved arrow labeled $\div 6$ points from the denominator 12 to the denominator 2.

Step One

Multiply the numerators and denominators.

Step Two

Simplify your answer if necessary.



And again...

$$\frac{5}{6} \times \frac{2}{3} = \frac{5 \times 2}{6 \times 3} = \frac{10}{18} = \frac{5}{9}$$

Step One

Multiply the numerators
and denominators.

$\div 2$

Step Two

Simplify your answer if
necessary.





Now it's time to
DIVIDE!



$$\frac{1}{2} \div \frac{1}{4} = \frac{1 \times 4}{2 \times 1} = \frac{4}{2} = \frac{2}{1} = 2$$

The diagram illustrates the division of fractions. It starts with $\frac{1}{2} \div \frac{1}{4}$. A white rounded rectangle highlights the first step: $\frac{1 \times 4}{2 \times 1}$. To the right, the fraction $\frac{4}{2}$ is shown with a curved arrow pointing from the numerator 4 to the denominator 2, labeled $\div 2$. Below $\frac{4}{2}$ is the fraction $\frac{2}{1}$, with a curved arrow pointing from the denominator 1 to the numerator 2, also labeled $\div 2$. The final result is $= 2$.

Step One

Keep, Switch and Flip.

Step Two

Simplify your answer if necessary.

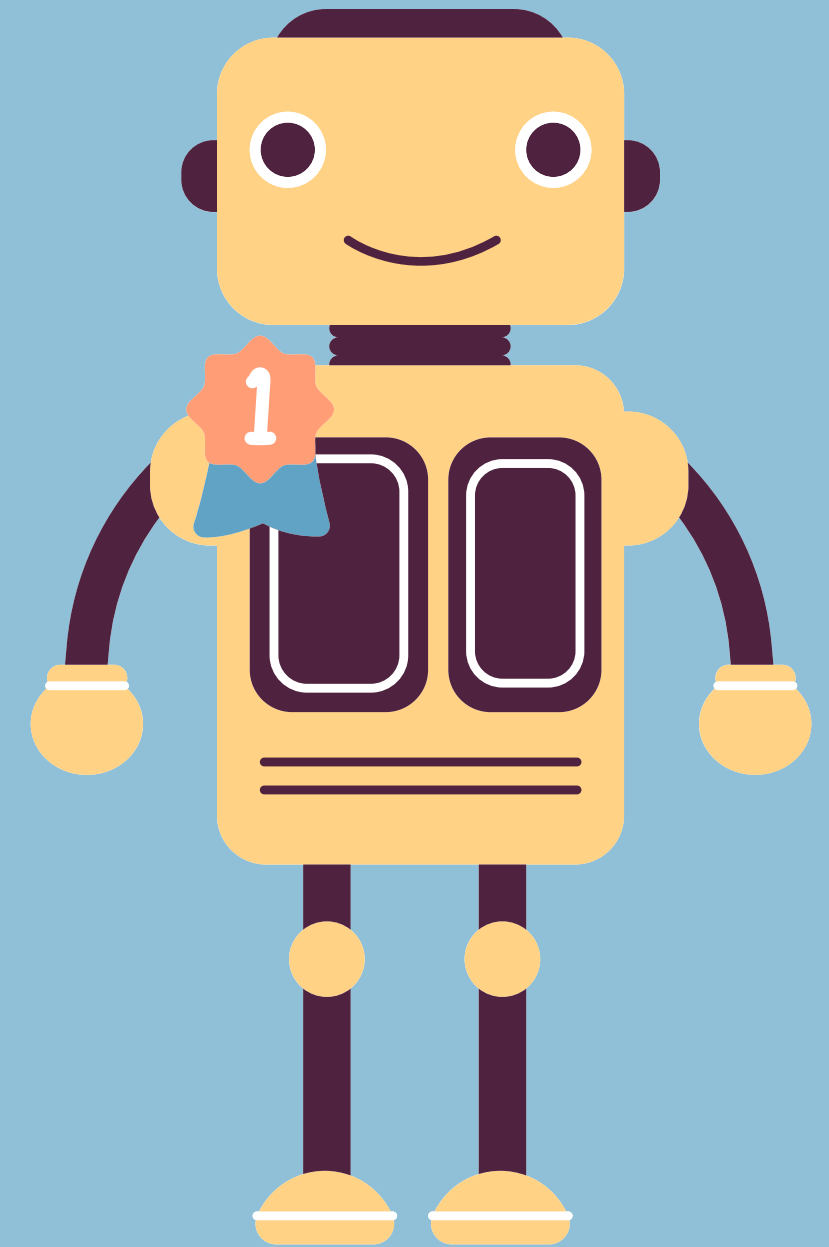
And again...

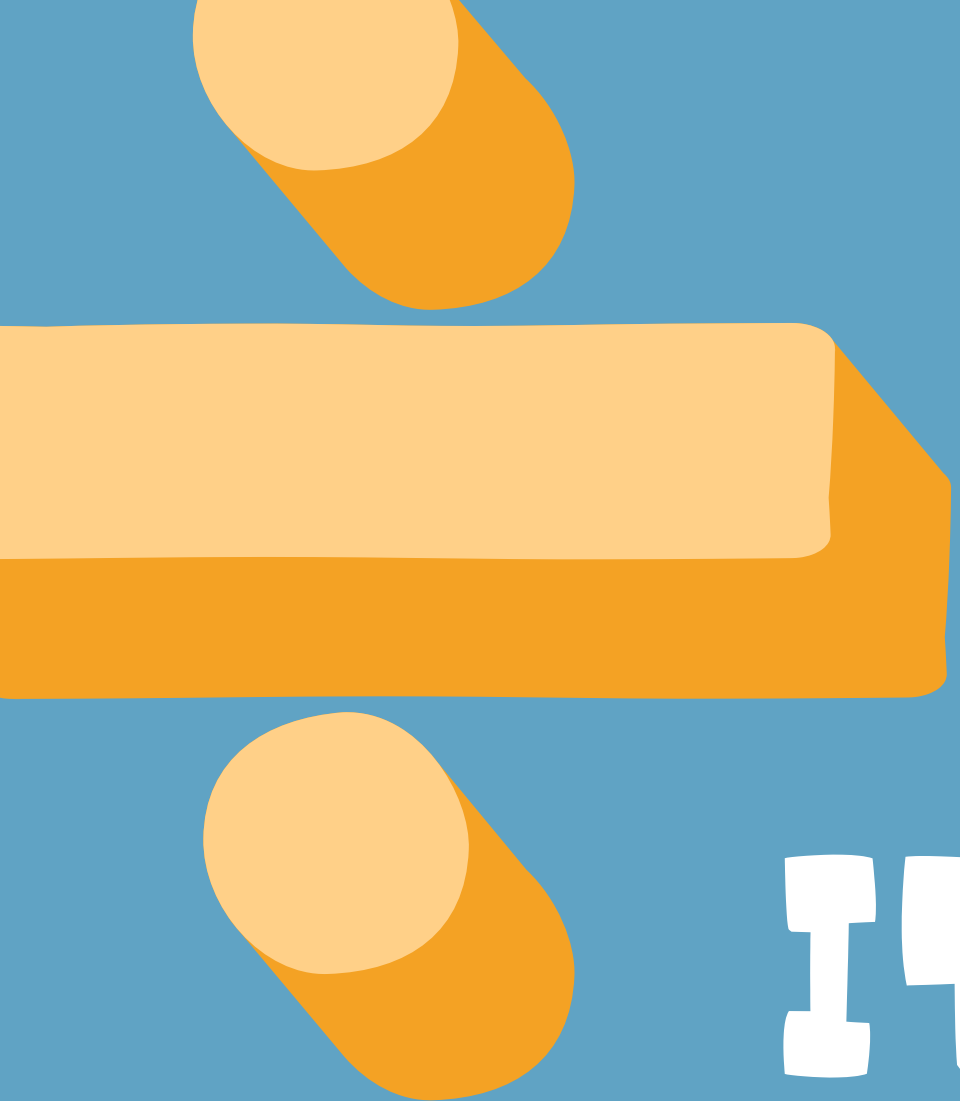
$$\frac{2}{5} \div \frac{2}{3} = \frac{2 \times 3}{5 \times 2} = \frac{6}{10} = \frac{3}{5}$$

The diagram illustrates the simplification of the fraction $\frac{6}{10}$ to $\frac{3}{5}$. A curved arrow labeled $\div 2$ points from the numerator 6 to the denominator 10, and another curved arrow labeled $\div 2$ points from the denominator 10 to the numerator 3, showing the division of both by 2.

Step One
Keep, Switch and Flip.

Step Two
Simplify your answer if necessary.





IT'S YOUR TURN!

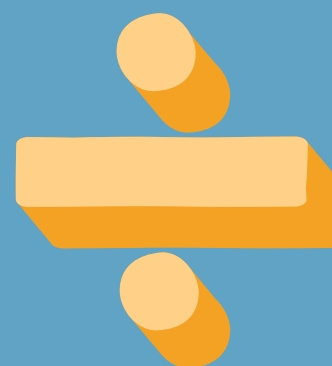
**Try the next three questions
in your workbook.**





QUESTION ONE

$$\frac{3}{4} \times \frac{1}{4}$$



QUESTION TWO

$$\frac{2}{9} \div \frac{1}{3}$$



QUESTION THREE

$$\frac{2}{6} \times \frac{2}{3}$$

ANSWERS

Q1

$$\frac{3}{16}$$

Q2

$$\frac{2}{3}$$

Q3

$$\frac{2}{9}$$



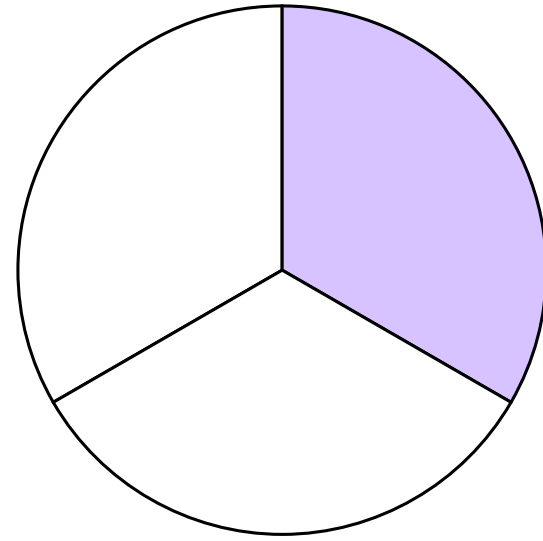
Equivalent Fractions



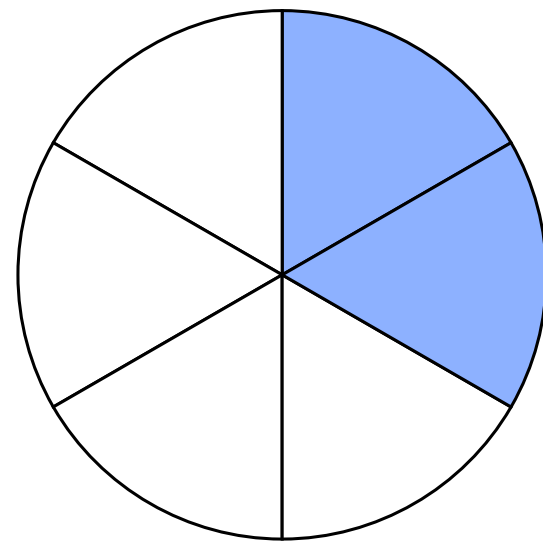
EXPLORE WRITING FRACTIONS AND UNDERSTANDING EQUIVALENT FRACTIONS

What are equivalent fractions?

Fractions that have the same value but with different numerators and denominators



=



$\frac{1}{3}$

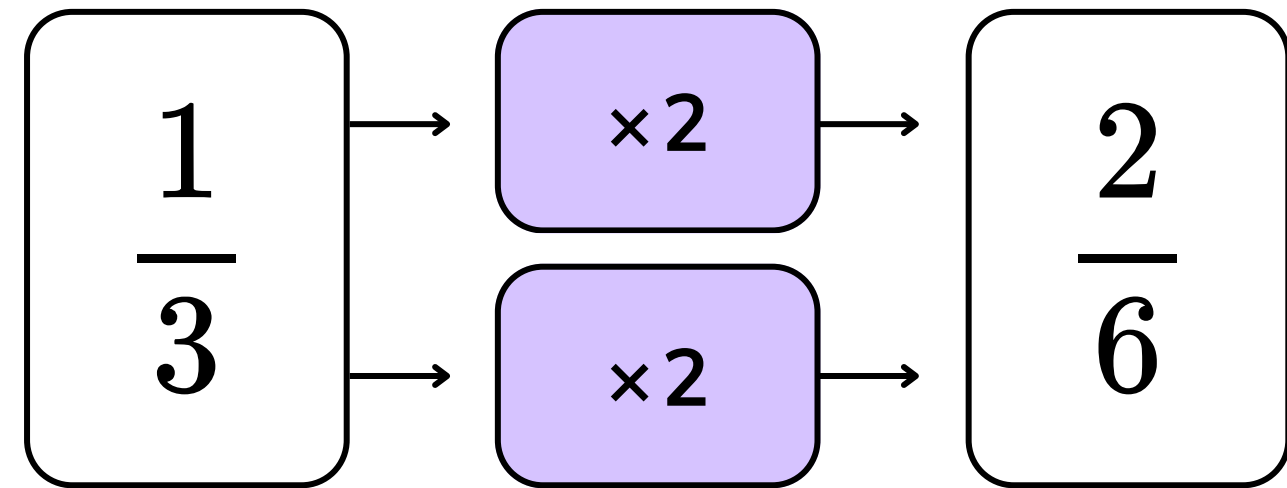
$\frac{2}{6}$

THEY
REPRESENT
THE SAME
PORTION
OF
A WHOLE

Finding Equivalent Fractions

To find equivalent fractions,
multiply or divide both the
numerator and denominator
by the same number.

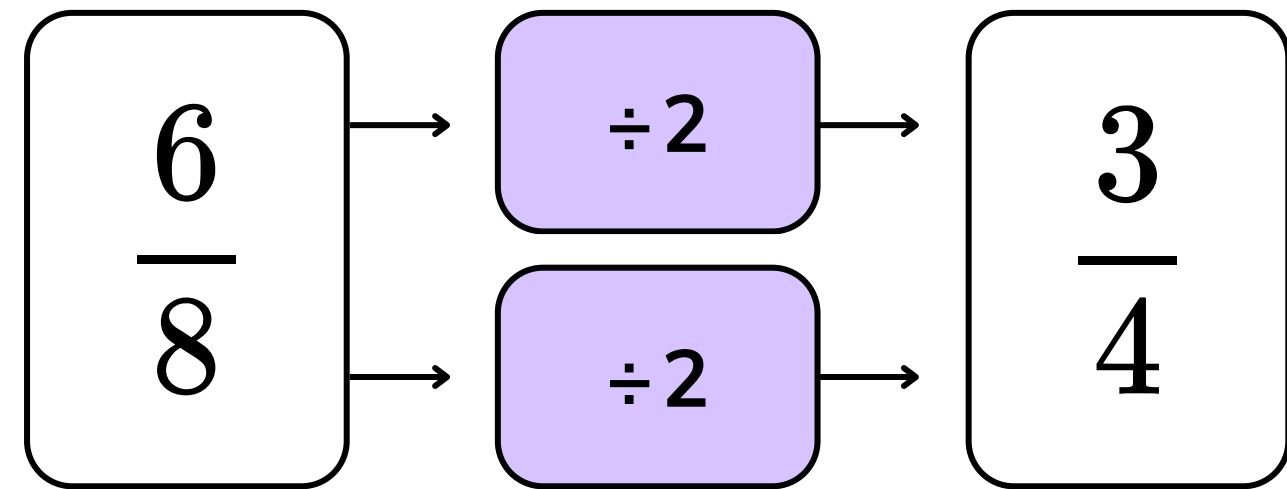
EXAMPLE 1



$\frac{1}{3}$ AND $\frac{2}{6}$ ARE
EQUIVALENT FRACTIONS

Finding Equivalent Fractions

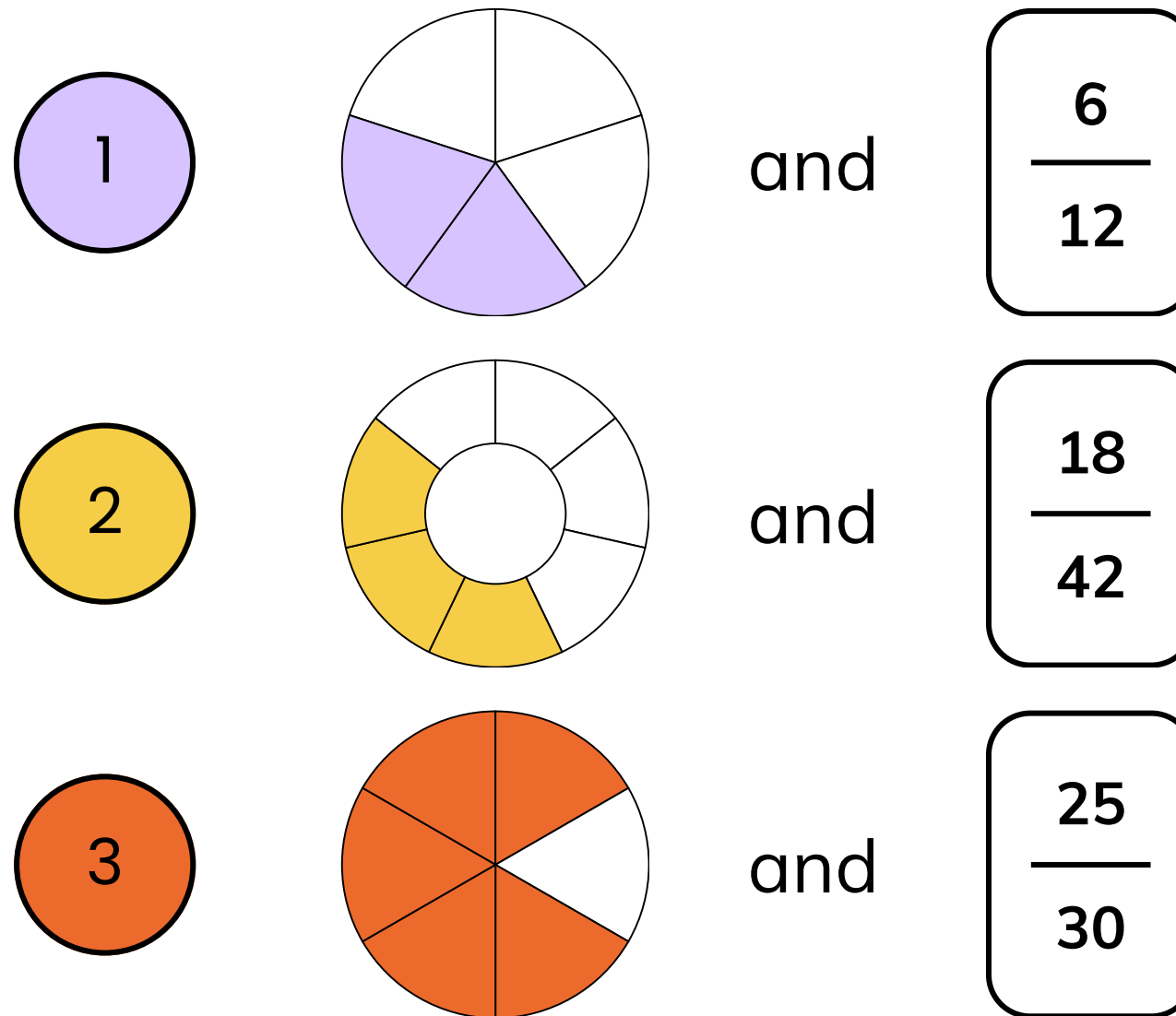
To find equivalent fractions,
multiply or **divide** both the
numerator and denominator
by the same number.



6/8 AND 3/4 ARE
EQUIVALENT FRACTIONS

Activity: Equivalent or not?

Can you tell if the fractions are equivalent or not?

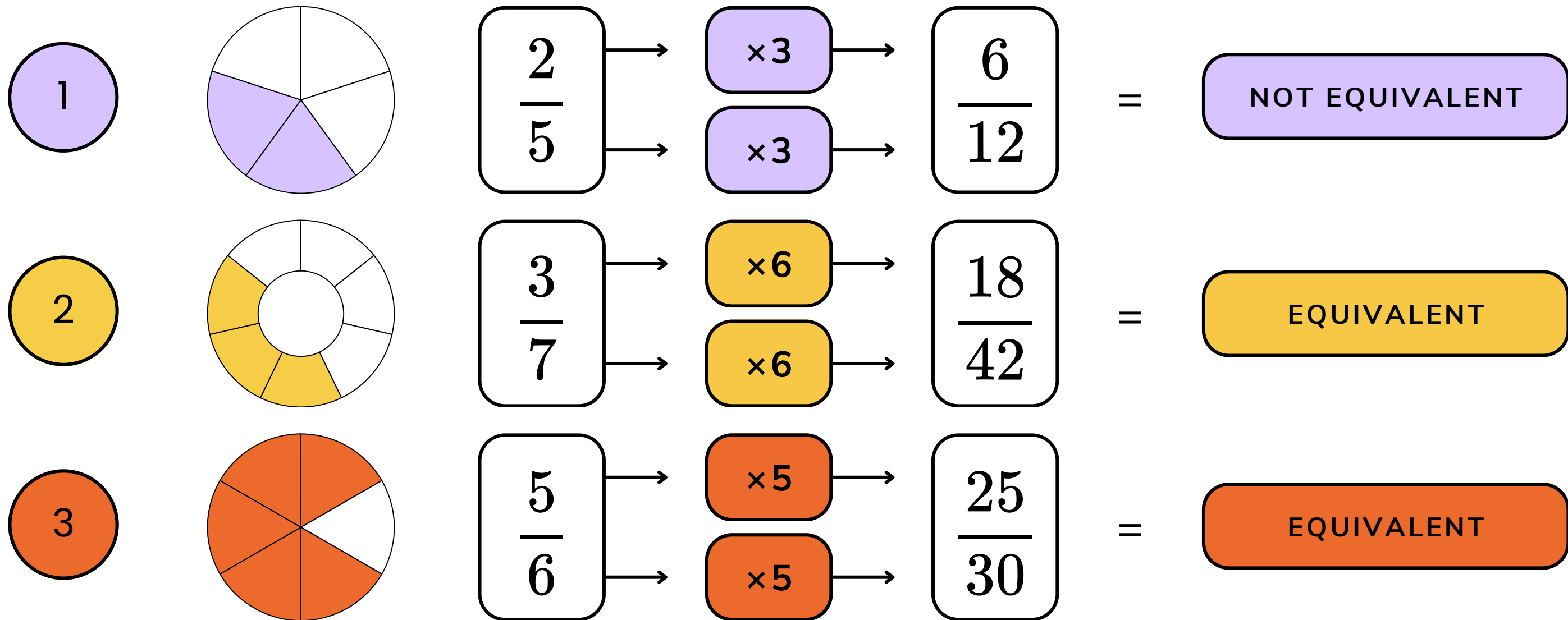


TIP

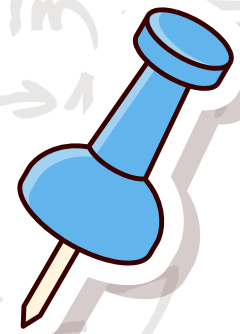
Think of a number that you can multiply to both the numerator and denominator of the first fraction.

Activity: Equivalent or not?

Can you tell if the fractions are equivalent or not?

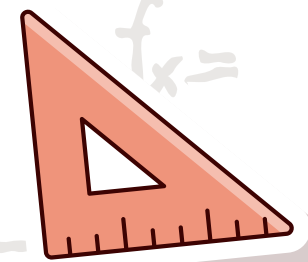


THANK
YOU



$$\frac{1}{4}$$

$$\frac{3}{7}$$



$$\frac{2}{3}$$

